

PHYS 143 – Mechanics and Thermal Physics

# Rotation of Rigid Bodies

*How do we describe and predict the motion of a spinning body, and what plays the role of velocity, mass, force, work and momentum for rotation?*

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By the end of this week, you should be able to:

1. Describe rotation using angular position, velocity and acceleration, and relate them to the linear motion of a point on the body.
2. Calculate moments of inertia and torques, and use  $\sum \tau = I\alpha$  to find angular accelerations.
3. Use work, power and energy methods for rotating and rolling bodies.
4. Use angular momentum and its conservation, including when the moment of inertia changes.

The detailed outcomes are in the checklist at the end of the lecture. **Reading:** Young and Freedman, Ch. 9, §9.1–9.6 and Ch. 10, §10.1, 10.2, 10.4–10.6.

Three contacts:

## 1 DESCRIBE

How do we describe the rotation of a rigid body?

## 2 MODEL

What is rotational inertia, what causes rotation, and what is the law for it?

## 3 APPLY

Can I use work, energy and angular momentum on rotating bodies?

Practice: concept check → worked example  
→ independent problem → application  
problem

## Where we start: a five-minute check

**Not marked. Answer individually, then we review.**

1. Find the circumference of a circle of radius 0.40 m, and the angle in degrees of one quarter of the circle.
2. A car starts from rest and accelerates uniformly at  $2.0 \text{ m/s}^2$  for 5.0 s. Find its final speed and the distance covered.
3. A 0.50 kg stone moves in a horizontal circle of radius 0.80 m at 4.0 m/s. Find its acceleration and the net force, with their direction.
4. Vectors  $\vec{A}$  (3.0 units) and  $\vec{B}$  (4.0 units) are  $30^\circ$  apart. Find the magnitude of  $\vec{A} \times \vec{B}$  and say how its direction is found.
5. A 20 kg crate is pushed 3.0 m by a 40 N force in 6.0 s. Find the work done and the average power.
6. A 2.0 kg ball falls from rest through 5.0 m. Use energy to find its speed on landing.

1.  $C = 2\pi r = 2\pi(0.40) = 2.5$  m; a quarter of a circle is  $90^\circ$ .
2.  $v = at = (2.0)(5.0) = 10$  m/s;  
 $x = \frac{1}{2}at^2 = 25$  m.
3.  $a = v^2/r = 16/0.80 = 20$  m/s<sup>2</sup> toward the centre;  $F = ma = 10$  N toward the centre.
4.  $|\vec{A} \times \vec{B}| = AB \sin 30^\circ = 6.0$  units; the direction is perpendicular to both vectors, by the right-hand rule.
5.  $W = Fs = (40)(3.0) = 120$  J;  
 $P = W/t = 20$  W.
6.  $v = \sqrt{2gh} = \sqrt{2(9.80)(5.0)} = 9.9$  m/s.

Week 1 (the cross product), Week 2 (constant acceleration), Week 3 (circular motion), Week 4 (work and power) and Week 5 (energy) are all reused this week. Kepler's laws, which were part of the old Lecture 8, belong to the Gravitation lecture.

## Contact 1: Describe

*How do we describe the rotation of a rigid body?*

## Weeks 2 to 5

a particle with a position  $x$ , velocity  $v$ , acceleration  $a$ .

*Where is it, how fast, how does that change?*

## Week 8

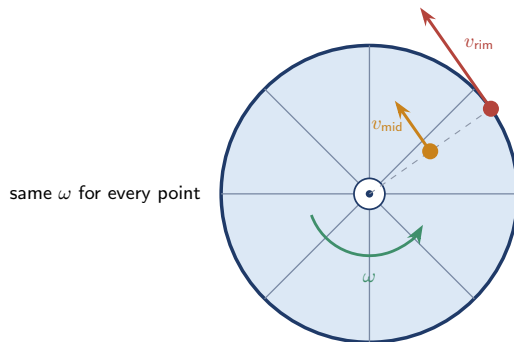
a *rigid body* (tyre, CD, fan, turbine) turning about an axis. Every point moves in a circle about the axis.

*What single description covers all of its points?*

### Opening question

A bicycle wheel of radius 0.35 m turns at 120 revolutions per minute. A point on the rim and a point halfway from the axle to the rim are marked. Which point has the greater *angular* velocity? Which has the greater *speed*, and by what factor?

## Two points on one wheel



### Predict first

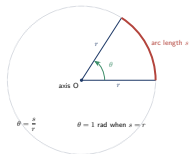
Sketch the wheel and the velocity of each point. Predict: the ratio of the two speeds, and the speed of the rim in m/s (a brisk walk is about 1.5 m/s).

# Defining the vocabulary of rotation

| Quantity                             | What it is                                                  | What it measures                                 | SI unit            | Representation                               |
|--------------------------------------|-------------------------------------------------------------|--------------------------------------------------|--------------------|----------------------------------------------|
| <b>Rigid body</b>                    | an object whose parts keep fixed distances from one another | a body that turns as a whole about an axis       | —                  | a shape with a marked axis                   |
| <b>Angular position</b> $\theta$     | the angle turned from a reference line                      | how far the body has rotated                     | rad                | arc length over radius, $s/r$                |
| <b>Angular velocity</b> $\omega$     | rate of change of $\theta$                                  | how fast, and in which sense, the body rotates   | rad/s              | slope of $\theta(t)$ ; vector along the axis |
| <b>Angular acceleration</b> $\alpha$ | rate of change of $\omega$                                  | how quickly the rotation speeds up or slows down | rad/s <sup>2</sup> | slope of $\omega(t)$                         |

**Interpret:**  $\omega$  answers *how fast, and which way, is it turning?* Every point of the body has the *same*  $\omega$  and  $\alpha$ .

# Angle in radians



## Rule: angle in radians

$$\theta = \frac{s}{r} \quad 1 \text{ rev} = 2\pi \text{ rad} = 360^\circ$$

$s$  is the arc length on a circle of radius  $r$ .

$$1 \text{ rad} = 57.3^\circ$$

$$1 \text{ rev/s} = 2\pi \text{ rad/s}$$

$$1 \text{ rpm} = \frac{2\pi}{60} \text{ rad/s} = 0.105 \text{ rad/s}$$

## Physical meaning

A radian is a ratio of two lengths and has no dimension; “rad” is a label for the unit used. Every formula that links an angle to a length needs **radians**.

### Decide first, using the wheel on the previous slides

1. Do all points of a turning wheel have the same angular velocity? The same speed?
2. A wheel turns at constant  $\omega$ . Is any point of it accelerating?
3. Can  $\omega$  be zero at an instant while  $\alpha$  is not?
4. Why do we measure angles in radians?

### Decide first, using the wheel on the previous slides

1. Do all points of a turning wheel have the same angular velocity? The same speed?
2. A wheel turns at constant  $\omega$ . Is any point of it accelerating?
3. Can  $\omega$  be zero at an instant while  $\alpha$  is not?
4. Why do we measure angles in radians?

### Answers

1. **Same  $\omega$ ; speed grows with the distance from the axis.**
2. **Yes.** Every point moves in a circle, so it has  $a_{\text{rad}} = \omega^2 r$  toward the axis.
3. **Yes.** A flywheel that is momentarily at rest while reversing.
4. **A radian is a ratio of lengths,  $s/r$ , so  $s = r\theta$  and  $v = r\omega$  hold without extra factors.**

## Rule: angular velocity and angular acceleration

$$\omega_{\text{av}} = \frac{\Delta\theta}{\Delta t} \quad \omega = \frac{d\theta}{dt} \quad \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

Units:  $\omega$  in rad/s,  $\alpha$  in rad/s<sup>2</sup>. For rotation about a fixed axis choose a positive sense (anticlockwise is positive):  $\omega$  and  $\alpha$  are then signed numbers.

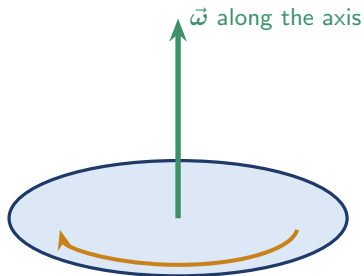
## Physical meaning

$\omega$  is the slope of  $\theta(t)$ .  $\alpha$  is the slope of  $\omega(t)$ . This is Week 2 with  $x \rightarrow \theta$ ,  
 $v \rightarrow \omega$ ,  $a \rightarrow \alpha$ .

## Common error

$\omega > 0$  and  $\alpha < 0$  together mean the body is *slowing down* while still turning in the positive sense, not that it has stopped.

# The direction of rotation: the right-hand rule



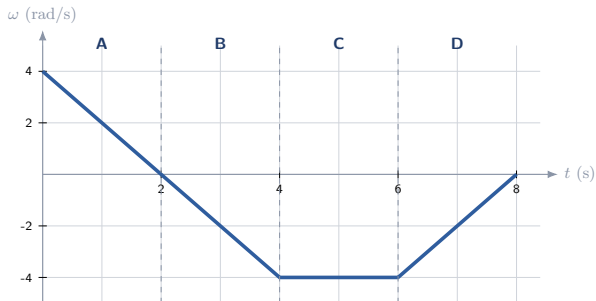
Right-hand rule: curl the fingers in the sense of rotation, the thumb gives the direction of  $\vec{\omega}$ .

Speeding up:  $\vec{\alpha}$  along  $\vec{\omega}$ . Slowing down:  $\vec{\alpha}$  opposite to  $\vec{\omega}$ .

rotation: anticlockwise seen from above

For a fixed axis the vector  $\vec{\omega}$  lies along the axis, so a sign is enough: positive for anticlockwise as seen from the tip of the axis vector, negative for clockwise.

## Concept check: reading an $\omega(t)$ graph



A wheel's angular velocity is plotted in four segments, A to D (anticlockwise is positive).

**Answer for each segment, with a reason**

- (a) The sense of rotation. (b) The sign of  $\alpha$ . (c) Is the wheel speeding up or slowing down? (d) The angle turned in segment A.

## Reading an $\omega(t)$ graph: answers

| Segment   | Sense ( $\omega$ ) | $\alpha$ (slope)       | Speed        | $\Delta\theta$ (area) |
|-----------|--------------------|------------------------|--------------|-----------------------|
| A (0–2 s) | anticlockwise      | $-2.0 \text{ rad/s}^2$ | slowing down | $+4.0 \text{ rad}$    |
| B (2–4 s) | clockwise          | $-2.0 \text{ rad/s}^2$ | speeding up  | $-4.0 \text{ rad}$    |
| C (4–6 s) | clockwise          | 0                      | constant     | $-8.0 \text{ rad}$    |
| D (6–8 s) | clockwise          | $+2.0 \text{ rad/s}^2$ | slowing down | $-4.0 \text{ rad}$    |

### Rule of thumb

$\omega$  and  $\alpha$  with the *same* sign: speeding up. *Opposite* signs: slowing down. The angle turned is the area under the  $\omega$ – $t$  graph; the net angle over 8 s is  $-12 \text{ rad}$  (clockwise).

## Example 1: from $\theta(t)$ to $\omega$ and $\alpha$

### Question

A turbine disc turns through an angle  $\theta(t) = 3.0t^2 - 0.20t^3$  ( $\theta$  in radians,  $t$  in seconds) while it is being shut down. Find (a)  $\theta$ ,  $\omega$  and  $\alpha$  at  $t = 2.0$  s; (b) the time at which  $\omega$  is greatest, and its value; (c) the average angular velocity between  $t = 2.0$  s and  $t = 4.0$  s.

### Predict first

Sketch  $\omega(t)$  before calculating. Does  $\omega$  increase all the time? What is  $\alpha$  at the instant  $\omega$  is greatest? Will  $\omega_{\text{av}}$  for the interval be the mean of the two end values?

## From $\theta(t)$ to $\omega$ and $\alpha$ : solution

**Set up:** differentiate:  $\omega = \frac{d\theta}{dt} = 6.0t - 0.60t^2$  and  $\alpha = \frac{d\omega}{dt} = 6.0 - 1.2t$ .

## From $\theta(t)$ to $\omega$ and $\alpha$ : solution

**Set up:** differentiate:  $\omega = \frac{d\theta}{dt} = 6.0t - 0.60t^2$  and  $\alpha = \frac{d\omega}{dt} = 6.0 - 1.2t$ .

**(a)** At  $t = 2.0$  s:  $\theta = 3.0(4.0) - 0.20(8.0) = 10.4$  rad;  $\omega = 12.0 - 2.4 = 9.6$  rad/s;  
 $\alpha = 6.0 - 2.4 = 3.6$  rad/s<sup>2</sup>.

## From $\theta(t)$ to $\omega$ and $\alpha$ : solution

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 $\alpha = 6.0 - 2.4 = 3.6$  rad/s<sup>2</sup>.

**(b)**  $\omega$  is greatest where  $\alpha = 0$ :  $6.0 - 1.2t = 0 \Rightarrow t = 5.0$  s, and  $\omega_{\text{max}} = 30 - 15 = 15$  rad/s.

## From $\theta(t)$ to $\omega$ and $\alpha$ : solution

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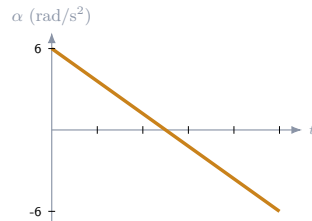
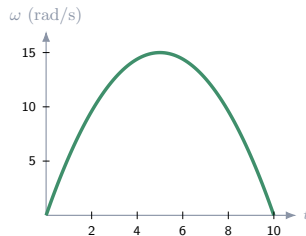
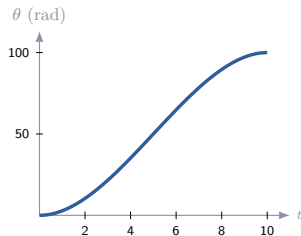
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**(b)**  $\omega$  is greatest where  $\alpha = 0$ :  $6.0 - 1.2t = 0 \Rightarrow t = 5.0$  s, and  $\omega_{\max} = 30 - 15 = 15$  rad/s.

**(c)**  $\theta(4.0) = 48.0 - 12.8 = 35.2$  rad, so  $\omega_{\text{av}} = \frac{35.2 - 10.4}{2.0} = 12.4$  rad/s.

**Evaluate:**  $\omega(2.0) = 9.6$  and  $\omega(4.0) = 14.4$  rad/s; their mean is 12.0 rad/s, not 12.4: the average is not the mean because  $\alpha$  is not constant. The disc stops at  $t = 10$  s ( $\omega = 0$ ). ✓

# The three graphs of $\theta(t)$



$\omega$  is the slope of the left graph;  $\alpha$  is the slope of the middle graph.  $\omega$  peaks where  $\alpha$  crosses zero ( $t = 5.0$  s), and  $\omega = 0$  where  $\theta$  stops increasing ( $t = 10$  s). Between 0 and 5 s,  $\omega$  and  $\alpha$  have the same sign (speeding up); after 5 s the signs differ (slowing down).

# Constant angular acceleration

## Rule: constant angular acceleration

$$\omega = \omega_0 + \alpha t \quad \theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0) \quad \theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t$$

valid **only** when  $\alpha$  is constant, with  $\theta$  in radians.

| Week 2 (linear)                      | Week 8 (rotation)                                         | Variable missing |
|--------------------------------------|-----------------------------------------------------------|------------------|
| $v = v_0 + at$                       | $\omega = \omega_0 + \alpha t$                            | $\theta$         |
| $x - x_0 = v_0 t + \frac{1}{2} at^2$ | $\theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2$ | $\omega$         |
| $v^2 = v_0^2 + 2a(x - x_0)$          | $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$      | $t$              |
| $x - x_0 = \frac{1}{2}(v_0 + v)t$    | $\theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t$     | $\alpha$         |

## EQUATION – angular velocity, acceleration

$$\omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt}$$

**Meaning.** the rate of turning, and the rate at which the rate of turning changes.

**Variables.**  $\theta(t)$  angle in radians;  $t$  time.

**Units.** rad/s and rad/s<sup>2</sup>.

**Interpretation.** slopes of  $\theta(t)$  and  $\omega(t)$ ; same signs, speeding up.

**Example.**  $\theta = 3.0t^2 - 0.20t^3$ :  $\omega = 9.6$  rad/s at 2.0 s.

## EQUATION – constant $\alpha$

$$\omega = \omega_0 + \alpha t$$

$$\theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2$$

**Meaning.** the Week 2 equations with  $x \rightarrow \theta$ ,  $v \rightarrow \omega$ ,  $a \rightarrow \alpha$ .

**Variables.**  $\omega_0$  initial angular velocity;  $\alpha$  constant.

**Units.** rad/s; rad.

**Interpretation.** valid only if  $\alpha$  is constant.

**Example.** the washing-machine drum.

### TOOLKIT 1 – Angular quantities and constant angular acceleration

1. **SKETCH** the body and the axis; choose the positive sense (anticlockwise); mark the signs of  $\omega_0$ ,  $\omega$  and  $\alpha$ .
  2. **UNITS:** convert degrees, revolutions and rpm to radians and rad/s *before* using any formula.
  3. **LIST** what is given and wanted among  $\theta - \theta_0$ ,  $\omega_0$ ,  $\omega$ ,  $\alpha$ ,  $t$ .
  4. **CHOOSE:** if  $\theta(t)$  is given, differentiate. If  $\alpha$  is constant, pick the equation that does *not* contain the quantity you neither know nor want.
  5. **SOLVE**, then convert radians to revolutions if asked ( $1 \text{ rev} = 2\pi \text{ rad}$ ).
- RULE:** the constant- $\alpha$  equations fail if  $\alpha$  changes with time.
- WATCH OUT:** using rpm or degrees in an equation; a negative  $\alpha$  for slowing down in the positive sense.
- CHECK:** does the sign of  $\alpha$  agree with speeding up or slowing down? Is the number of revolutions reasonable?

## Example 2: a braking washing-machine drum

### Question

A washing-machine drum spinning at 900 rpm is braked uniformly to rest in 12 s. Find (a) its angular acceleration, and (b) the number of revolutions it makes while stopping.

### Predict first

Sketch  $\omega(t)$ . Will  $\alpha$  be positive or negative? At 900 rpm = 15 rev/s for 12 s the drum would make 180 revolutions at constant speed: will the actual number be more or less than that?

**Set up:** anticlockwise positive;  $\omega_0 = 900 \times \frac{2\pi}{60} = 94.2 \text{ rad/s}$ ,  $\omega = 0$ ,  $t = 12 \text{ s}$ ;  $\alpha$  constant.

## The braking drum: solution

**Set up:** anticlockwise positive;  $\omega_0 = 900 \times \frac{2\pi}{60} = 94.2 \text{ rad/s}$ ,  $\omega = 0$ ,  $t = 12 \text{ s}$ ;  $\alpha$  constant.

**(a)**  $\omega = \omega_0 + \alpha t \Rightarrow \alpha = \frac{0 - 94.2}{12} = -7.85 \text{ rad/s}^2$ .

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**(a)**  $\omega = \omega_0 + \alpha t \Rightarrow \alpha = \frac{0 - 94.2}{12} = -7.85 \text{ rad/s}^2$ .

**(b)** The equation without  $\alpha$ :  $\theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t = \frac{1}{2}(94.2)(12) = 565 \text{ rad}$ , which is  $\frac{565}{2\pi} = 90.0$  revolutions.

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**Evaluate:**  $\alpha < 0$  as predicted (slowing in the positive sense). The drum makes 90 revolutions, half of 180, because the average angular velocity of a uniform slowing is half the initial value. Check with  $\theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2 = 1131 - 565 = 565 \text{ rad}$ . ✓

### Independent problem

A centrifuge rotor accelerates uniformly from rest to 3000 rpm in 40.0 s. Find (a) its angular acceleration, (b) the number of revolutions it makes in the 40.0 s, and (c) the time it takes to complete its first 100 revolutions.

### Predict first

Sketch  $\omega(t)$  and mark the area for part (b). For (c): the angle grows as  $t^2$ . Will the time for 100 revolutions be more or less than 10 s (a quarter of 40 s)?

**Set up:**  $\omega = 3000 \times \frac{2\pi}{60} = 314.2 \text{ rad/s}$  at  $t = 40.0 \text{ s}$ ;  $\omega_0 = 0$ .

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**(a)**  $\alpha = \frac{\omega}{t} = \frac{314.2}{40.0} = 7.85 \text{ rad/s}^2$ .

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**(a)**  $\alpha = \frac{\omega}{t} = \frac{314.2}{40.0} = 7.85 \text{ rad/s}^2$ .

**(b)**  $\theta = \frac{1}{2}(\omega_0 + \omega)t = \frac{1}{2}(314.2)(40.0) = 6283 \text{ rad} = 1000 \text{ revolutions}$ .

**Set up:**  $\omega = 3000 \times \frac{2\pi}{60} = 314.2 \text{ rad/s}$  at  $t = 40.0 \text{ s}$ ;  $\omega_0 = 0$ .

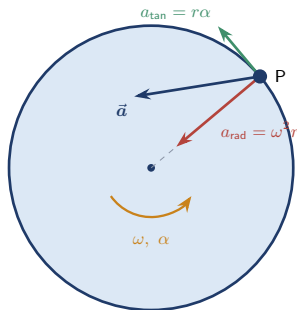
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(c)  $100 \text{ rev} = 628.3 \text{ rad} = \frac{1}{2}\alpha t^2 \Rightarrow t = \sqrt{\frac{2(628.3)}{7.85}} = 12.6 \text{ s}$ .

**Evaluate:**  $12.6 \text{ s} > 10 \text{ s}$  as predicted. Check:  $\theta \propto t^2$ , so 100 rev is  $\frac{1}{10}$  of 1000 rev and  $t = 40.0/\sqrt{10} = 12.6 \text{ s}$ . ✓

## From the body to a point on it



**Rule: a point at distance  $r$  from the axis**

$$s = r\theta \quad v = r\omega$$

$$a_{\text{tan}} = r\alpha \quad a_{\text{rad}} = \omega^2 r = \frac{v^2}{r}$$

with  $\theta, \omega, \alpha$  in **radians**.

### Physical meaning

Every point has the same  $\omega$  and  $\alpha$ ; the speed and the accelerations grow in proportion to  $r$ .  $a_{\text{tan}}$  changes the speed;  $a_{\text{rad}}$  (Week 3, centripetal acceleration) changes the direction. The magnitude of the total acceleration is  $a = \sqrt{a_{\text{tan}}^2 + a_{\text{rad}}^2}$ .

## EQUATION – point on a rotating body

$$s = r\theta, \quad v = r\omega, \quad a_{\text{tan}} = r\alpha, \quad a_{\text{rad}} = \omega^2 r$$

**Meaning.** the linear motion of a point is fixed by the angular motion of the body and its distance from the axis.

**Variables.**  $r$  distance from the axis;  $\theta$ ,  $\omega$ ,  $\alpha$  in radians.

**Units.** m, m/s, m/s<sup>2</sup>.

**Interpretation.** points far from the axis move faster;  $a_{\text{tan}}$  changes the speed,  $a_{\text{rad}}$  the direction.

**Example.** the wheel of radius 0.45 m after 3.0 s.

## Predict first

A wheel turns at constant  $\omega$ .  
Predict the ratio of the speeds of two points at  $r$  and  $2r$  before calculating.

### TOOLKIT 2 – Linking linear and angular motion

1. **DECIDE** whether the question asks about the body ( $\omega, \alpha, \theta$ ) or about a point on it ( $v, a, s$ ).
2. **RADIUS:** the distance  $r$  of that point from the *axis* (not from the centre of the body).
3. **RADIANS:**  $\omega$  in rad/s,  $\theta$  in rad, before using  $v = r\omega$  or  $s = r\theta$ .
4. **COMPONENTS:**  $a_{\text{tan}} = r\alpha$  along the motion,  $a_{\text{rad}} = \omega^2 r$  toward the axis; add them as perpendicular vectors.

**RULE:** if  $\omega$  is constant,  $a_{\text{tan}} = 0$ , but  $a_{\text{rad}} = \omega^2 r$  is still present.

**WATCH OUT:** using rpm in  $v = r\omega$ ; forgetting  $a_{\text{rad}}$  for a point on a spinning wheel.

**CHECK:** is  $a_{\text{rad}} = v^2/r$ ? Does  $v$  grow in proportion to  $r$ ?

## Example 3: a point on a speeding-up wheel

### Question

A wheel of radius  $0.45\text{ m}$  starts from rest and speeds up with a constant angular acceleration of  $2.4\text{ rad/s}^2$ . At  $t = 3.0\text{ s}$  find, for a point on the rim, (a) the angular velocity, (b) the speed, (c) the tangential and radial accelerations, and (d) the magnitude of the total acceleration and its angle with the radius.

### Predict first

Sketch the point and its two acceleration components. At that instant, which component is the larger, and is the total close to one of them?

## A point on a speeding-up wheel: solution

**Set up:**  $\alpha = 2.4 \text{ rad/s}^2$  (constant),  $\omega_0 = 0$ ,  $r = 0.45 \text{ m}$ ,  $t = 3.0 \text{ s}$ .

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**(a)**  $\omega = \alpha t = 7.2 \text{ rad/s}$ .     **(b)**  $v = r\omega = (0.45)(7.2) = 3.24 \text{ m/s}$ .

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**(c)**  $a_{\text{tan}} = r\alpha = (0.45)(2.4) = 1.08 \text{ m/s}^2$ ;  $a_{\text{rad}} = \omega^2 r = (7.2)^2(0.45) = 23.3 \text{ m/s}^2$ .

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**(d)**  $a = \sqrt{1.08^2 + 23.3^2} = 23.4 \text{ m/s}^2$ ; the angle with the inward radius is  $\arctan \frac{1.08}{23.3} = 2.7^\circ$ .

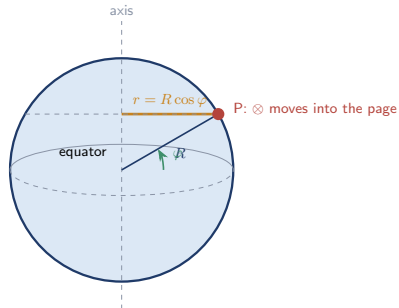
**Evaluate:**  $a_{\text{rad}} \gg a_{\text{tan}}$  at this instant, so the total points almost at the centre, as predicted.

Check:  $a_{\text{rad}} = v^2/r = 3.24^2/0.45 = 23.3 \text{ m/s}^2$ . ✓

## Your turn: the speed of the ground beneath you

### Independent problem

The Earth turns once in 24.0 h (a simplification: the true period is 23.93 h). Accra is at latitude  $5.6^\circ$  N. The Earth's radius is  $6.37 \times 10^6$  m. Find (a) the Earth's angular velocity, (b) the speed of a person in Accra because of the rotation, and (c) the person's centripetal acceleration as a fraction of  $g$ .



### Predict first

Sketch the circle of latitude: what is its radius for a point near the equator? Is the speed closer to 4.6 m/s, 46 m/s or 460 m/s?

**Set up:** the point moves on a circle of latitude of radius  
 $r = R \cos \varphi = (6.37 \times 10^6) \cos 5.6^\circ = 6.34 \times 10^6 \text{ m}.$

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$$\text{(a)} \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{(24.0)(3600)} = 7.27 \times 10^{-5} \text{ rad/s.}$$

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$$\text{(a)} \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{(24.0)(3600)} = 7.27 \times 10^{-5} \text{ rad/s}.$$

$$\text{(b)} \quad v = r\omega = (6.34 \times 10^6)(7.27 \times 10^{-5}) = 461 \text{ m/s}.$$

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$$\text{(b)} \quad v = r\omega = (6.34 \times 10^6)(7.27 \times 10^{-5}) = 461 \text{ m/s.}$$

$$\text{(c)} \quad a_{\text{rad}} = \omega^2 r = (7.27 \times 10^{-5})^2 (6.34 \times 10^6) = 0.0335 \text{ m/s}^2, \text{ which is } \frac{0.0335}{9.80} = 3.4 \times 10^{-3} g.$$

**Evaluate:** the answer is of order 460 m/s, faster than sound in air, yet we feel nothing because the air and the ground move with us at constant velocity, and  $a_{\text{rad}}$  is only about 0.3% of  $g$ . ✓

## Common mistakes and the correct habit

### Mistake

Using degrees or rev/min in  $v = r\omega$  or  $s = r\theta$ .

Measuring  $r$  from the centre of the body.

Forgetting  $a_{\text{rad}}$  for a point on a wheel at constant  $\omega$ .

Using the constant- $\alpha$  equations when  $\alpha$  varies.

Reading  $\omega > 0$ ,  $\alpha < 0$  as stopped.

### Instead

Convert to radians and rad/s first.

$r$  is the distance from the axis.

$a_{\text{rad}} = \omega^2 r$  is still present.

Differentiate or integrate instead.

It means slowing while still turning.

## Contact 1: what we can now do

**Angle,  $\omega$ ,  $\alpha$**  (radians, signs)



**Constant  $\alpha$**  (Week 2 equations)



**A point on the body**  $v = r\omega$ ,  $a_{\text{tan}} = r\alpha$ ,  
 $a_{\text{rad}} = \omega^2 r$

### You should be able to

- convert between degrees, radians, revolutions and rpm;
- find  $\omega$  and  $\alpha$  from  $\theta(t)$ , and read the graphs;
- use the constant- $\alpha$  equations when  $\alpha$  is constant;
- relate the motion of a point to that of the body.

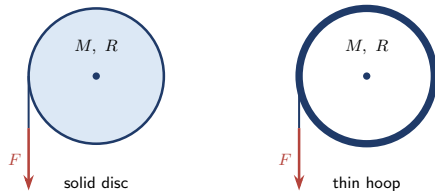
### Connect: Lecture 11

The same angular quantities describe a spinning molecule: the rotational kinetic energy of gas molecules enters the heat capacity.

## Contact 2: Model

*What is rotational inertia, what causes rotation, and what is the law for it?*

## An opening question: the disc and the hoop



### Poll

A cord is wound on the rim of a solid disc and of a thin hoop. Both have the same mass  $M$  and radius  $R$ , both are free to turn about their centres, and the cord is pulled with the same force  $F$ . Which one reaches the larger angular acceleration, or are they equal?

### Predict first

Same force, same lever arm. What property of the body, other than its mass, must matter? Sketch where the mass of each body sits.

# Defining the vocabulary of rotational dynamics

| Quantity                                                                                                                                           | What it is                                   | What it measures                                   | SI unit         | Representation                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------|-----------------|---------------------------------------------------|
| <b>Moment of inertia</b> $I$                                                                                                                       | the sum $\sum m_i r_i^2$ about a chosen axis | how strongly the body resists a change of rotation | $\text{kg m}^2$ | the way the mass is spread about the axis         |
| <b>Torque</b> $\tau$                                                                                                                               | the turning effect of a force about an axis  | how effectively a force turns the body             | $\text{N m}$    | force $\times$ lever arm; a vector along the axis |
| <b>Rotational kinetic energy</b>                                                                                                                   | the energy of the turning motion             | the work needed to set the body turning            | $\text{J}$      | $\frac{1}{2} I \omega^2$                          |
| <b>Interpret:</b> $I$ plays the role of mass and torque the role of force: the question is <i>which axis, and how is the mass spread about it?</i> |                                              |                                                    |                 |                                                   |

# Kinetic energy of rotation and the moment of inertia

Treat the body as many small particles. Particle  $i$  has mass  $m_i$  at distance  $r_i$  from the axis, so its speed is  $v_i = r_i\omega$ :

$$K = \sum_i \frac{1}{2} m_i v_i^2 = \sum_i \frac{1}{2} m_i (r_i \omega)^2 = \frac{1}{2} \left( \sum_i m_i r_i^2 \right) \omega^2.$$

**Rule: kinetic energy of rotation and moment of inertia**

$$K = \frac{1}{2} I \omega^2 \quad I = \sum_i m_i r_i^2 \quad [I] = \text{kg m}^2$$

$r_i$  is the distance from the **axis of rotation**, so  $I$  depends on the axis chosen.

## Physical meaning

$I$  plays the role of mass for rotation: it measures how hard it is to change the rotation. Mass far from the axis counts more, in proportion to  $r^2$ .

## Common error

$I$  is not a property of the body alone. The same body has different values of  $I$  about different axes.

## Centre of mass of particles on a line

$$x_{\text{cm}} = \frac{m_1x_1 + m_2x_2 + \cdots}{m_1 + m_2 + \cdots} = \frac{\sum m_i x_i}{M}$$

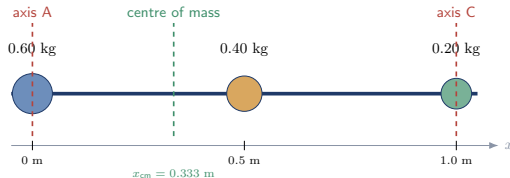
The weighted average position of the mass: the point where the body balances. For a uniform body of symmetric shape it lies at the centre of symmetry.

We need only this definition today, to compare axes. (The centre of mass, and how it moves, are treated in more detail in the supplementary notes.) Two particles of 2.0 kg at  $x = 0$  and 6.0 kg at  $x = 4.0$  m give  $x_{\text{cm}} = \frac{0 + (6.0)(4.0)}{8.0} = 3.0$  m: it lies nearer the heavier particle.

## Example 4: one body, three axes

### Question

Three small masses are fixed on a light rigid rod (negligible mass) as shown. Find  $I$  for an axis perpendicular to the rod through (a) mass A, (b) mass C, and (c) the centre of mass. Then find  $K$  for each axis when  $\omega = 6.0 \text{ rad/s}$ .



### Predict first

Sketch the three axes. About which axis do you expect the smallest  $I$ , and about which the largest? Is  $I$  about C close to  $I$  about A?

## One body, three axes: solution

**Set up:**  $M = 1.20 \text{ kg}$ ;  $x_{\text{cm}} = \frac{(0.60)(0) + (0.40)(0.50) + (0.20)(1.00)}{1.20} = 0.333 \text{ m}$ .

$I = \sum m_i r_i^2$  with  $r_i$  measured from each axis.

## One body, three axes: solution

**Set up:**  $M = 1.20 \text{ kg}$ ;  $x_{\text{cm}} = \frac{(0.60)(0) + (0.40)(0.50) + (0.20)(1.00)}{1.20} = 0.333 \text{ m}$ .

$I = \sum m_i r_i^2$  with  $r_i$  measured from each axis.

**(a) Axis A:**  $I_A = 0 + (0.40)(0.50)^2 + (0.20)(1.00)^2 = 0.30 \text{ kg m}^2$ .

**(b) Axis C:**  $I_C = (0.60)(1.00)^2 + (0.40)(0.50)^2 + 0 = 0.70 \text{ kg m}^2$ .

**(c) Centre of mass:**  $I_{\text{cm}} = (0.60)(0.333)^2 + (0.40)(0.167)^2 + (0.20)(0.667)^2 = 0.167 \text{ kg m}^2$ .

## One body, three axes: solution

**Set up:**  $M = 1.20 \text{ kg}$ ;  $x_{\text{cm}} = \frac{(0.60)(0) + (0.40)(0.50) + (0.20)(1.00)}{1.20} = 0.333 \text{ m}$ .

$I = \sum m_i r_i^2$  with  $r_i$  measured from each axis.

**(a) Axis A:**  $I_A = 0 + (0.40)(0.50)^2 + (0.20)(1.00)^2 = 0.30 \text{ kg m}^2$ .

**(b) Axis C:**  $I_C = (0.60)(1.00)^2 + (0.40)(0.50)^2 + 0 = 0.70 \text{ kg m}^2$ .

**(c) Centre of mass:**  $I_{\text{cm}} = (0.60)(0.333)^2 + (0.40)(0.167)^2 + (0.20)(0.667)^2 = 0.167 \text{ kg m}^2$ .

**Kinetic energy**  $K = \frac{1}{2} I \omega^2$  at  $6.0 \text{ rad/s}$ :  $5.4 \text{ J}$  (A),  $12.6 \text{ J}$  (C),  $3.0 \text{ J}$  (centre of mass).

**Evaluate:**  $I$  is smallest about the centre of mass and largest about the far end, as predicted.

Pattern:  $I_A - I_{\text{cm}} = 0.133 = M(0.333)^2$  and  $I_C - I_{\text{cm}} = 0.533 = M(0.667)^2$ .

# The parallel-axis theorem

## Rule: parallel-axis theorem

$$I = I_{\text{cm}} + Md^2$$

$I_{\text{cm}}$  is the moment of inertia about the axis through the centre of mass;  $d$  is the distance between that axis and the parallel axis;  $M$  is the total mass.

## Check on the example:

$$I_A = 0.167 + (1.20)(0.333)^2 = 0.300$$

$$I_C = 0.167 + (1.20)(0.667)^2 = 0.700 \quad \checkmark$$

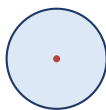
## Physical meaning

For a given direction of the axis,  $I$  is smallest about the axis through the centre of mass. Moving the axis away can only increase  $I$ . (The proof is in the supplementary notes.)

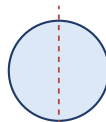
# Moments of inertia of standard bodies



thin hoop, centre axis  
 $I = MR^2$



solid disc, centre axis  
 $I = \frac{1}{2}MR^2$



solid sphere, diameter axis  
 $I = \frac{2}{5}MR^2$



thin shell, diameter axis  
 $I = \frac{2}{3}MR^2$



thin rod, centre axis  
 $I = \frac{1}{12}ML^2$



thin rod, end axis  
 $I = \frac{1}{3}ML^2$

For the same  $M$  and  $R$ : hoop  $>$  disc, and shell  $>$  solid sphere. The more mass sits far from the axis, the larger the  $I$ . The integrals that give these results are an unassessed extension in the supplementary notes.

## EQUATION – moment of inertia and

$K_{\text{rot}}$

$$I = \sum_i m_i r_i^2, \quad K = \frac{1}{2} I \omega^2$$

**Meaning.** how the mass is distributed about the axis; the energy of the turning.

**Variables.**  $m_i$  masses;  $r_i$  their perpendicular distances from the axis.

**Units.**  $\text{kg m}^2$ ; J.

**Interpretation.** mass far from the axis counts as  $r^2$ ;  $I$  depends on the axis.

**Example.** the rod and spheres:  
 $I = 0.204 \text{ kg m}^2$ .

## EQUATION – parallel-axis theorem

$$I = I_{\text{cm}} + M d^2$$

**Meaning.** the moment of inertia about an axis parallel to one through the centre of mass.

**Variables.**  $I_{\text{cm}}$  about the centre of mass;  $M$  total mass;  $d$  separation of the axes.

**Units.**  $\text{kg m}^2$ .

**Interpretation.**  $I$  is smallest about the axis through the centre of mass.

**Example.**

$$0.167 + 1.20(0.333)^2 = 0.300 \text{ kg m}^2.$$

### TOOLKIT 3 – Moment of inertia

1. **AXIS FIRST:** mark the axis on the sketch;  $I$  has no meaning without it.
  2. **CHOOSE THE METHOD:** separate particles:  $\sum mr^2$ . A standard body about its central axis: the table. A parallel axis:  $I_{\text{cm}} + Md^2$ . A composite body: add the  $I$  of the parts about the *same* axis.
  3. **DISTANCES:**  $r$  is the perpendicular distance to the axis, not to the origin or the centre.
  4. **CALCULATE** and then  $K = \frac{1}{2}I\omega^2$  if energy is asked.
- RULE:** moments of inertia about one axis add; they do not add across different axes.
- WATCH OUT:** using  $MR^2$  for every body; using the table value for an axis that is not the central axis;  $d$  measured from the wrong point.
- CHECK:** is  $I$  between the values for the most compact and the most spread-out distribution of the same mass?

## Example 5: a composite body

### Question

A uniform thin rod of mass  $0.50\text{ kg}$  and length  $0.60\text{ m}$  carries a small sphere of mass  $0.40\text{ kg}$  at each end (treat each sphere as a point mass). Find  $I$  for an axis perpendicular to the rod through (a) its centre and (b) one end.

### Predict first

Sketch the rod with the two axes. By what factor, roughly, will  $I$  about the end exceed  $I$  about the centre: 1, 2 or 4?

## Example 5: a composite body

### Question

A uniform thin rod of mass 0.50 kg and length 0.60 m carries a small sphere of mass 0.40 kg at each end (treat each sphere as a point mass). Find  $I$  for an axis perpendicular to the rod through (a) its centre and (b) one end.

### Predict first

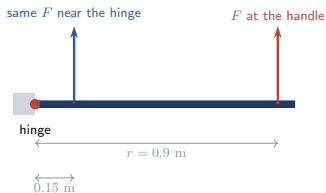
Sketch the rod with the two axes. By what factor, roughly, will  $I$  about the end exceed  $I$  about the centre: 1, 2 or 4?

**(a) Centre:** rod  $\frac{1}{12}ML^2 = \frac{1}{12}(0.50)(0.60)^2 = 0.015$ ; spheres  $2(0.40)(0.30)^2 = 0.072$ ;  
 $I = 0.087 \text{ kg m}^2$ .

**(b) End:** rod  $\frac{1}{3}ML^2 = 0.060$ ; sphere on the axis 0; far sphere  $(0.40)(0.60)^2 = 0.144$ ;  
 $I = 0.204 \text{ kg m}^2$ .

**Evaluate:** ratio 2.3, close to 2 as predicted. Check by the theorem:  
 $0.087 + (1.30)(0.30)^2 = 0.204$ . ✓

# Torque: why a door opens at the handle



The same force  $F$  is applied perpendicular to the door, once at the handle ( $0.90\text{ m}$  from the hinge) and once  $0.15\text{ m}$  from the hinge.

*Which opens the door faster? What if  $\vec{F}$  pointed along the door, toward the hinge?*

## Predict first

Rank the **turning effects** of these cases before you read on: the handle push, the push near the hinge, the push along the door.

## Concept check: torque before the formula

### Decide first

1. Can a force that acts on a body produce no torque?
2. Can the net force on a body be zero while the net torque is not?
3. Does a force of the same size always give the same torque about a given axis?

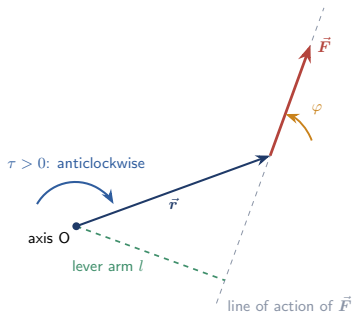
## Concept check: torque before the formula

### Decide first

1. Can a force that acts on a body produce no torque?
2. Can the net force on a body be zero while the net torque is not?
3. Does a force of the same size always give the same torque about a given axis?

### Answers

1. **Yes.** If its line of action passes through the axis.
2. **Yes.** Equal and opposite forces on opposite sides of the axis (a couple) turn the body without moving its centre.
3. **No.** Torque depends on where and at what angle the force acts:  $\tau = rF \sin \varphi$ .



## Rule: torque about an axis O

$$\tau = rF \sin \varphi = F_{\perp} r = Fl$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$\varphi$  is the angle between  $\vec{r}$  and  $\vec{F}$ ;  
 $l = r \sin \varphi$  is the lever arm. Unit: N m.

## Physical meaning

Torque is the turning effect of a force about a chosen axis: it grows with the force and with the perpendicular distance from the axis to the *line of action*. Sign:  $\tau > 0$  for anticlockwise,  $\tau < 0$  for clockwise. It is zero if the line of action passes through the axis ( $\varphi = 0^\circ$  or  $180^\circ$ ).

# Torque and energy are different quantities

## Common error

Torque is measured in N m and work in joules (J). The units have the same dimensions, but they are different quantities: write torque as N m and energy as J. The direction of  $\vec{\tau}$  is along the axis (right-hand rule); energy has no direction.

## Net torque

For several forces about the same axis, add the *signed* torques:

$$\sum \tau = \tau_1 + \tau_2 + \cdots$$

A zero net *force* does not imply a zero net *torque*.

The door: handle push  $\tau = (0.90)F$ ; near the hinge  $\tau = (0.15)F$ , six times smaller; along the door,  $\varphi = 0$  and  $\tau = 0$ .

## EQUATION – torque

$$\tau = rF \sin \varphi = F_{\perp} r = Fl, \quad \vec{\tau} = \vec{r} \times \vec{F}$$

**Meaning.** the turning effect of a force about a chosen axis.

**Variables.**  $r$  from the axis to the point of application;  $\varphi$  angle between  $\vec{r}$  and  $\vec{F}$ ;  $l = r \sin \varphi$  lever arm.

**Units.** N m (not J).

**Interpretation.** sign from the sense of turning; zero if the line of action passes through the axis.

**Example.** the door:  $0.90F$  at the handle,  $0.15F$  near the hinge.

## Predict first

Predict the ranking of a push at 0.90 m, a push at 0.15 m, and a push along the door, all of equal force.

### TOOLKIT 4 – Torque

1. **AXIS:** mark the axis O, the point where each force acts, and each line of action.
  2. **MAGNITUDE:**  $\tau = rF \sin \varphi$ , with  $\varphi$  the angle between  $\vec{r}$  and  $\vec{F}$ ; or  $F_{\perp} r$ ; or  $F l$  with the lever arm. Pick whichever is easiest from the sketch.
  3. **SIGN:** anticlockwise positive, clockwise negative, decided from the sketch before calculating.
  4. **NET TORQUE:** add the signed torques about the *same* axis; a force acting at the axis, or with its line of action through it, gives zero.
- RULE:** only the component of the force perpendicular to  $\vec{r}$  gives a torque.
- WATCH OUT:** using the angle between  $\vec{F}$  and the horizontal instead of between  $\vec{F}$  and  $\vec{r}$ ; mixing axes.
- CHECK:** does the sign agree with the direction the force would turn the body?

**A wrench turns a bolt. Forces (a) to (d) each have magnitude 20 N. Rank their torques about the bolt.**

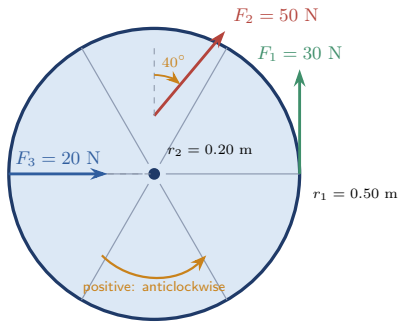
- (a) Perpendicular to the handle, at 0.30 m from the bolt.
- (b) At  $30^\circ$  to the handle, at 0.30 m.
- (c) Perpendicular to the handle, at 0.15 m.
- (d) Along the line of the handle, pointing at the bolt.
- (e) Two forces of 10 N in opposite directions, each perpendicular to the ends of a 0.40 m diameter, applied to a wheel: what are the net force and the net torque?

A wrench turns a bolt. Forces (a) to (d) each have magnitude 20 N. Rank their torques about the bolt.

- (a) Perpendicular to the handle, at 0.30 m from the bolt.
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- (c) Perpendicular to the handle, at 0.15 m.
- (d) Along the line of the handle, pointing at the bolt.
- (e) Two forces of 10 N in opposite directions, each perpendicular to the ends of a 0.40 m diameter, applied to a wheel: what are the net force and the net torque?

(a)  $20(0.30) = 6.0 \text{ N m}$ ; (b)  $20(0.30) \sin 30^\circ = 3.0 \text{ N m}$ ; (c)  $20(0.15) = 3.0 \text{ N m}$ ; (d) 0.  
Ranking: (a) > (b) = (c) > (d). (e) The net force is zero, but the net torque is  $2(10)(0.20) = 4.0 \text{ N m}$ : a **couple** turns the wheel without accelerating its centre.

## Example 6: a wheel with three forces



### Question

Three forces act on a wheel that turns about its axle through O. Find the torque of each about O and the net torque, and say which way the wheel tends to turn.

### Predict first

Sketch the lever arm of each force. Which forces turn the wheel anticlockwise, which clockwise, and which have zero torque?

## The wheel with three forces: solution

$F_1$ : tangent to the rim, so  $\varphi = 90^\circ$ :  $\tau_1 = +r_1 F_1 = +(0.50)(30) = +15.0 \text{ N m}$  (anticlockwise).

## The wheel with three forces: solution

$F_1$ : tangent to the rim, so  $\varphi = 90^\circ$ :  $\tau_1 = +r_1 F_1 = +(0.50)(30) = +15.0 \text{ N m}$  (anticlockwise).

$F_2$ :  $\varphi = 40^\circ$  between  $\vec{r}_2$  and  $\vec{F}_2$ , and it turns the wheel clockwise:

$$\tau_2 = -r_2 F_2 \sin 40^\circ = -(0.20)(50)(0.643) = -6.43 \text{ N m}.$$

## The wheel with three forces: solution

$F_1$ : tangent to the rim, so  $\varphi = 90^\circ$ :  $\tau_1 = +r_1 F_1 = +(0.50)(30) = +15.0 \text{ N m}$  (anticlockwise).

$F_2$ :  $\varphi = 40^\circ$  between  $\vec{r}_2$  and  $\vec{F}_2$ , and it turns the wheel clockwise:

$$\tau_2 = -r_2 F_2 \sin 40^\circ = -(0.20)(50)(0.643) = -6.43 \text{ N m}.$$

$F_3$ : its line of action passes through O, so  $\tau_3 = 0$ .

## The wheel with three forces: solution

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$$\tau_2 = -r_2 F_2 \sin 40^\circ = -(0.20)(50)(0.643) = -6.43 \text{ N m}.$$

$F_3$ : its line of action passes through O, so  $\tau_3 = 0$ .

**Net torque:**  $\sum \tau = +15.0 - 6.43 + 0 = +8.6 \text{ N m}$ : the wheel tends to turn **anticlockwise**.

**Evaluate:** the signs agree with the sketch. The axle supplies a force that has no torque about O, so it does not appear in  $\sum \tau$  even though  $\sum \vec{F} \neq 0$ . ✓

## Newton's second law for rotation

Consider particle  $i$  of a rigid body turning about a fixed axis. The tangential force on it gives  $F_{\text{tan},i} = m_i a_{\text{tan},i} = m_i r_i \alpha$ , and its torque is

$$\tau_i = r_i F_{\text{tan},i} = m_i r_i^2 \alpha.$$

Add all the particles. The internal forces between particles cancel in pairs, and so do their torques, leaving only the external torques:

### Rule: Newton's second law for rotation

$$\sum \tau_{\text{ext}} = I \alpha$$

about a fixed axis, with  $\tau$  and  $I$  taken about the **same** axis.  $\alpha$  is in  $\text{rad/s}^2$ .

### Physical meaning

Compare  $\sum F = ma$ : torque plays the role of force and  $I$  that of mass. For a given torque, a larger  $I$  gives a smaller  $\alpha$ .

### EQUATION – Newton's second law for rotation

$$\sum \tau_{\text{ext}} = I\alpha$$

**Meaning.** the net external torque about an axis produces an angular acceleration, inversely proportional to  $I$ .

**Variables.**  $\sum \tau_{\text{ext}}$  net torque;  $I$  moment of inertia, both about the *same* axis;  $\alpha$  angular acceleration.

**Units.**  $\text{N m} = \text{kg m}^2 \text{ rad/s}^2$ .

**Interpretation.** the rotational analogue of  $\sum F = ma$ .

**Example.** the flywheel:  $24/1.5 = 16 \text{ rad/s}^2$ .

### Predict first

The same torque acts on a solid disc and a hoop of equal mass and radius. Predict which has the larger  $\alpha$ .

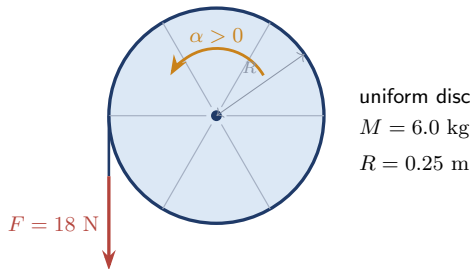
### ✂ TOOLKIT 5 – Rotational dynamics, $\sum \tau = I\alpha$

1. **SKETCH:** mark the axis and the positive sense of rotation; draw a free-body diagram of *each* body, with the point where each force acts.
  2. **TORQUES:**  $\sum \tau$  about the axis (Toolkit 4), with signs;  $I$  about the same axis (Toolkit 3).
  3. **WRITE**  $\sum \tau = I\alpha$  for the rotating body.
  4. **ATTACHED OBJECTS:**  $\sum F = ma$  for each one, plus the cord condition  $a = R\alpha$  (the cord does not slip or stretch).
  5. **SOLVE**, then use Toolkit 1 for  $\omega$  and  $\theta$ .
- RULE:** each rotating body gets its own equation; forces through the axis give no torque.
- WATCH OUT:** assuming the tension equals the weight; assuming the same tension on both sides of a pulley with mass; using the wrong sign for the sense of rotation.
- CHECK:** does  $\alpha$  fall when  $I$  rises? Is  $T < mg$  for a hanging mass that accelerates down?

## Example 7: pulling a cord wound on a disc

### Question

A uniform disc ( $M = 6.0 \text{ kg}$ ,  $R = 0.25 \text{ m}$ ) turns freely about its axle. A light cord wound round its rim is pulled with a constant tangential force of  $18 \text{ N}$ . Starting from rest, find (a)  $\alpha$ , (b)  $\omega$  and the angle turned after  $2.0 \text{ s}$ . (c) What would  $\alpha$  be for a thin hoop of the same  $M$  and  $R$ ?



### Predict first

Sketch the torque and the sign of  $\alpha$ . Will the hoop's  $\alpha$  be larger, smaller or the same as the disc's?

## The disc and the cord: solution

**Set up:** anticlockwise positive.  $I = \frac{1}{2}MR^2 = \frac{1}{2}(6.0)(0.25)^2 = 0.1875 \text{ kg m}^2$ .  
 $\tau = FR = (18)(0.25) = 4.50 \text{ N m}$ .

## The disc and the cord: solution

**Set up:** anticlockwise positive.  $I = \frac{1}{2}MR^2 = \frac{1}{2}(6.0)(0.25)^2 = 0.1875 \text{ kg m}^2$ .

$$\tau = FR = (18)(0.25) = 4.50 \text{ N m}.$$

$$\text{(a)} \quad \alpha = \frac{\tau}{I} = \frac{4.50}{0.1875} = 24.0 \text{ rad/s}^2.$$

**Set up:** anticlockwise positive.  $I = \frac{1}{2}MR^2 = \frac{1}{2}(6.0)(0.25)^2 = 0.1875 \text{ kg m}^2$ .

$$\tau = FR = (18)(0.25) = 4.50 \text{ N m}.$$

$$\text{(a)} \quad \alpha = \frac{\tau}{I} = \frac{4.50}{0.1875} = 24.0 \text{ rad/s}^2.$$

$$\text{(b)} \quad \omega = \alpha t = 48.0 \text{ rad/s}; \quad \theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(24.0)(2.0)^2 = 48.0 \text{ rad (7.6 revolutions)}.$$

## The disc and the cord: solution

**Set up:** anticlockwise positive.  $I = \frac{1}{2}MR^2 = \frac{1}{2}(6.0)(0.25)^2 = 0.1875 \text{ kg m}^2$ .  
 $\tau = FR = (18)(0.25) = 4.50 \text{ N m}$ .

**(a)**  $\alpha = \frac{\tau}{I} = \frac{4.50}{0.1875} = 24.0 \text{ rad/s}^2$ .

**(b)**  $\omega = \alpha t = 48.0 \text{ rad/s}$ ;  $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(24.0)(2.0)^2 = 48.0 \text{ rad}$  (7.6 revolutions).

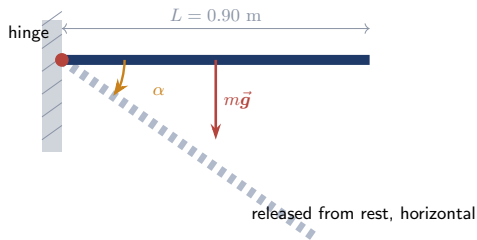
**(c)** Hoop:  $I = MR^2 = 0.375 \text{ kg m}^2$ , so  $\alpha = 4.50/0.375 = 12.0 \text{ rad/s}^2$ , **half** that of the disc.

**Evaluate:** the same torque gives the hoop the smaller  $\alpha$  because its mass is all at the rim, answering the opening question.  $K = \frac{1}{2}I\omega^2 = \frac{1}{2}(0.1875)(48.0)^2 = 216 \text{ J}$ , and  $\tau\theta = (4.50)(48.0) = 216 \text{ J}$ : preview of the next contact. ✓

## Your turn: a rod released from the horizontal

### Independent problem

A uniform rod ( $M = 1.2 \text{ kg}$ ,  $L = 0.90 \text{ m}$ ) is hinged at one end and released from rest in a horizontal position. At the instant of release find (a) its angular acceleration, and (b) the linear acceleration of the free end.



### Predict first

Which torque acts about the hinge, and where does the weight act? Is the acceleration of the free end larger or smaller than  $g$ ?

**Set up:** axis at the hinge; clockwise positive. The hinge force has no torque about the hinge, so only the weight (at the centre) contributes.  $I = \frac{1}{3}ML^2 = \frac{1}{3}(1.2)(0.90)^2 = 0.324 \text{ kg m}^2$ .

**Set up:** axis at the hinge; clockwise positive. The hinge force has no torque about the hinge, so only the weight (at the centre) contributes.  $I = \frac{1}{3}ML^2 = \frac{1}{3}(1.2)(0.90)^2 = 0.324 \text{ kg m}^2$ .

**(a)**  $\tau = Mg\frac{L}{2} = (1.2)(9.80)(0.45) = 5.29 \text{ N m}$ , so  $\alpha = \frac{\tau}{I} = \frac{5.29}{0.324} = 16.3 \text{ rad/s}^2$ .

**Set up:** axis at the hinge; clockwise positive. The hinge force has no torque about the hinge, so only the weight (at the centre) contributes.  $I = \frac{1}{3}ML^2 = \frac{1}{3}(1.2)(0.90)^2 = 0.324 \text{ kg m}^2$ .

(a)  $\tau = Mg\frac{L}{2} = (1.2)(9.80)(0.45) = 5.29 \text{ N m}$ , so  $\alpha = \frac{\tau}{I} = \frac{5.29}{0.324} = 16.3 \text{ rad/s}^2$ .

(b)  $a_{\text{tip}} = r\alpha = L\alpha = (0.90)(16.3) = 14.7 \text{ m/s}^2 = 1.5 g$ .

**Evaluate:** the free end accelerates *faster* than free fall: it is pulled down by the rest of the rod. Check: the point at  $\frac{2}{3}L = 0.60 \text{ m}$  has  $a = (0.60)(16.3) = 9.8 \text{ m/s}^2 = g$ . This holds only at the instant of release: as the rod turns the lever arm shrinks. ✓

### Common mistakes and the correct habit

#### Mistake

Taking  $I$  about the wrong axis, or using  $MR^2$  for every body.

Using the angle between  $\vec{F}$  and the horizontal for torque.

Writing torque in joules.

Assuming the tension equals the weight on a pulley with mass.

Assuming zero net force means zero net torque.

#### Instead

Name the axis; use the table or the parallel-axis theorem.

The angle is between  $\vec{r}$  and  $\vec{F}$ , or use the lever arm.

Torque is in N m; energy is in J.

Write  $\sum F = ma$  and  $\sum \tau = I\alpha$  with  $a = R\alpha$ .

A couple turns a body with no net force.

## Contact 2: what we can now do

**Moment of inertia**  $I = \sum mr^2$ ,  
 $K = \frac{1}{2}I\omega^2$

↓

**Torque**  $\tau = rF \sin \varphi$

↓

**Rotational dynamics**  $\sum \tau = I\alpha$

### You should be able to

- calculate  $I$  for particles, standard and composite bodies;
- calculate a torque and a net torque, with signs;
- use  $\sum \tau = I\alpha$  with a cord or a hinge;
- say how  $\alpha$  changes with  $\tau$  and  $I$ .

### Connect: Lectures 4, 5 and 7

Torque acting through an angle does work as force does through a distance; angular momentum is conserved as momentum is.

## Contact 3: Apply

*Can I use work, energy and angular momentum on rotating bodies?*

## Weeks 4 and 5

$$W_{\text{net}} = \Delta K \text{ and}$$

$$K_1 + U_1 + W_{\text{other}} = K_2 + U_2 \text{ for a particle.}$$

*Do they still work for a spinning body?*

## Week 8

a constant torque drives a flywheel for several seconds.

*How much work is done, and how fast is energy delivered?*

### Opening question

A motor applies a constant torque to a flywheel that starts from rest. Is the power delivered by the motor constant in time, increasing, or decreasing? Is the work done in the second half of the time equal to, more than, or less than the work done in the first half?

# Defining work, power and angular momentum

| Quantity                        | What it is                                             | What it measures                                | SI unit                  | Representation            |
|---------------------------------|--------------------------------------------------------|-------------------------------------------------|--------------------------|---------------------------|
| <b>Work of a torque</b>         | the energy a torque transfers as the body turns        | how much a torque changes the rotational energy | J                        | $\tau \Delta\theta$       |
| <b>Power</b>                    | the rate at which a torque delivers energy             | how fast energy is supplied to the rotation     | W                        | $\tau\omega$              |
| <b>Rolling without slipping</b> | a wheel's contact point is momentarily at rest         | the link between rolling and sliding speeds     | m/s                      | $v_{\text{cm}} = R\omega$ |
| <b>Angular momentum</b> $L$     | $I\omega$ , or $\vec{r} \times \vec{p}$ for a particle | the quantity of rotational motion about a point | $\text{kg m}^2/\text{s}$ | a vector along the axis   |

**Interpret:** the linear ideas return with  $F \rightarrow \tau$ ,  $x \rightarrow \theta$ ,  $v \rightarrow \omega$ ,  $m \rightarrow I$ ,  $p \rightarrow L$ .

## Work and power of a torque

A tangential force  $F$  at radius  $r$  moves its point of application through the arc  $s = r\theta$ , so the work is  $W = Fs = (Fr)\theta = \tau\theta$ .

**Rule: work, power and the work–energy theorem for rotation**

$$W = \int \tau d\theta = \tau \Delta\theta \text{ (constant } \tau) \quad P = \frac{dW}{dt} = \tau\omega$$

$$W_{\text{net}} = \Delta K_{\text{rot}} = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$$

for rotation about a fixed axis, with  $\Delta\theta$  in radians.

|          | motion                   | inertia | cause  | law                   | work, power              | $K$                    |
|----------|--------------------------|---------|--------|-----------------------|--------------------------|------------------------|
| linear   | $x, v, a$                | $m$     | $F$    | $\sum F = ma$         | $Fx, Fv$                 | $\frac{1}{2}mv^2$      |
| rotation | $\theta, \omega, \alpha$ | $I$     | $\tau$ | $\sum \tau = I\alpha$ | $\tau\theta, \tau\omega$ | $\frac{1}{2}I\omega^2$ |

## Example 8: a flywheel under a constant torque

### Question

A flywheel with  $I = 1.5 \text{ kg m}^2$  is at rest. A motor applies a constant torque of  $24 \text{ N m}$  for  $6.0 \text{ s}$ . Find (a)  $\alpha$ ,  $\omega$  and the angle turned, (b) the work done by the torque, checked against  $\Delta K$ , and (c) the average power and the instantaneous power at  $t = 6.0 \text{ s}$ .

### Predict first

Sketch  $P(t) = \tau\omega$  against  $t$ . Will the instantaneous power at  $6.0 \text{ s}$  be less than, equal to, or greater than the average power? By what factor?

## A flywheel under a constant torque: solution

(a)  $\alpha = \frac{\tau}{I} = \frac{24}{1.5} = 16.0 \text{ rad/s}^2$ ;  $\omega = \alpha t = 96 \text{ rad/s}$ ;  $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(16.0)(6.0)^2 = 288 \text{ rad}$   
(45.8 revolutions).

## A flywheel under a constant torque: solution

(a)  $\alpha = \frac{\tau}{I} = \frac{24}{1.5} = 16.0 \text{ rad/s}^2$ ;  $\omega = \alpha t = 96 \text{ rad/s}$ ;  $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(16.0)(6.0)^2 = 288 \text{ rad}$   
(45.8 revolutions).

(b)  $W = \tau\Delta\theta = (24)(288) = 6.9 \times 10^3 \text{ J}$ . Check:  $\Delta K = \frac{1}{2}(1.5)(96)^2 = 6.9 \times 10^3 \text{ J}$ . ✓

## A flywheel under a constant torque: solution

(a)  $\alpha = \frac{\tau}{I} = \frac{24}{1.5} = 16.0 \text{ rad/s}^2$ ;  $\omega = \alpha t = 96 \text{ rad/s}$ ;  $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(16.0)(6.0)^2 = 288 \text{ rad}$   
(45.8 revolutions).

(b)  $W = \tau\Delta\theta = (24)(288) = 6.9 \times 10^3 \text{ J}$ . Check:  $\Delta K = \frac{1}{2}(1.5)(96)^2 = 6.9 \times 10^3 \text{ J}$ . ✓

(c)  $P_{\text{av}} = \frac{W}{t} = \frac{6912}{6.0} = 1.15 \times 10^3 \text{ W}$ ;  $P(6.0 \text{ s}) = \tau\omega = (24)(96) = 2.30 \times 10^3 \text{ W}$ .

**Evaluate:**  $\omega$  grows linearly, so  $P = \tau\omega$  grows linearly: the final power is exactly twice the average, and more work is done in the second half than in the first (three times as much). Same pattern as Week 4 for a constant force on a body starting from rest. ✓

# Energy with rotation, and rolling

Both the work–energy theorem and the Week 5 energy equation hold, with the kinetic energy of the whole body:

**Rule: energy equation with rotation**

$$K_1 + U_1 + W_{\text{other}} = K_2 + U_2 \quad K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2$$

For **rolling without slipping**:  $v_{\text{cm}} = R\omega$ .

## Physical meaning

A rolling body has two kinetic energies: translation of the centre of mass and rotation about it. The two are linked by  $v_{\text{cm}} = R\omega$ , so the rotation takes a fixed share of the energy.

## Why friction does no work

The point of contact is momentarily at rest, so *static* friction does no work, and mechanical energy is conserved. If the body slips, kinetic friction acts and takes energy.

### EQUATION – work and power of a torque

$$W = \tau \Delta\theta, \quad P = \tau\omega$$

**Meaning.** the energy a torque delivers, and its rate.

**Variables.**  $\tau$  constant torque;  $\Delta\theta$  angle in radians;  $\omega$ .

**Units.** J and W.

**Interpretation.** a bigger  $\omega$  means more power for the same torque.

**Example.** the flywheel:  $6.9 \times 10^3$  J;  
 $2.30 \times 10^3$  W at the end.

### EQUATION – rolling kinetic energy

$$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2$$

**Meaning.** translation of the centre of mass plus rotation about it.

**Variables.**  $M$ ,  $v_{\text{cm}}$ ;  $I_{\text{cm}}$ ,  $\omega$ ; for rolling,  $v_{\text{cm}} = R\omega$ .

**Units.** J.

**Interpretation.** rotation takes a fixed share, so a rolling body is slower than a sliding one.

**Example.** the race down the incline.

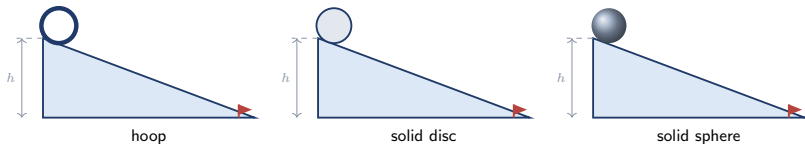
### TOOLKIT 6 – Work, power and energy in rotation

1. **STATES:** two states (1 and 2); give  $v$  or  $\omega$ , heights, and which parts rotate. Sketch the energy bars.
2. **KINETIC ENERGY:**  $K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2$  for each moving part; for rolling,  $\omega = v_{\text{cm}}/R$ .
3. **OTHER WORK:**  $W = \tau\Delta\theta$  for a torque;  $W = Fs$  for a force; power  $P = \tau\omega$ . Static friction in rolling does no work.
4. **WRITE**  $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$  and solve.
5. **CHOOSE:** energy for a speed between two states;  $\sum \tau = I\alpha$  for an acceleration, a time or a force.

**WATCH OUT:** leaving out the rotational term;  $\Delta\theta$  in degrees; a rope on a pulley: all the parts move, so the speed of the block and the speed of the rim must agree.

**CHECK:** is the speed of a rolling body smaller than for a frictionless slide? Does the second method agree?

## Concept check: a race down an incline



released together from rest at  $h = 0.80$  m; same  $M$  and  $R$ ; rolling without slipping

### Poll

A thin hoop, a solid disc and a solid sphere all have the same mass and radius. Released together from rest at the same height, they roll without slipping down an incline. Which reaches the bottom first, and does the mass or the radius matter?

### Predict first

Which body has the largest share of its energy in rotation? Rank the speeds at the bottom and compare them with a block sliding without friction.

## The race: solution

**Set up:**  $I = cMR^2$  with  $c = 1$  (hoop),  $\frac{1}{2}$  (disc),  $\frac{2}{5}$  (sphere). Static friction does no work, so  $Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$  with  $\omega = v/R$ :

$$Mgh = \frac{1}{2}Mv^2(1 + c) \quad \Rightarrow \quad v = \sqrt{\frac{2gh}{1 + c}}.$$

## The race: solution

**Set up:**  $I = cMR^2$  with  $c = 1$  (hoop),  $\frac{1}{2}$  (disc),  $\frac{2}{5}$  (sphere). Static friction does no work, so  $Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$  with  $\omega = v/R$ :

$$Mgh = \frac{1}{2}Mv^2(1 + c) \quad \Rightarrow \quad v = \sqrt{\frac{2gh}{1 + c}}.$$

**Execute** ( $h = 0.80$  m,  $2gh = 15.7$  m<sup>2</sup>/s<sup>2</sup>):

| Body                             | $c$  | $v$ at the bottom |
|----------------------------------|------|-------------------|
| frictionless block (no rotation) | 0    | 3.96 m/s          |
| solid sphere                     | 0.40 | 3.35 m/s          |
| solid disc                       | 0.50 | 3.23 m/s          |
| thin hoop                        | 1.00 | 2.80 m/s          |

## The race: solution

**Set up:**  $I = cMR^2$  with  $c = 1$  (hoop),  $\frac{1}{2}$  (disc),  $\frac{2}{5}$  (sphere). Static friction does no work, so  $Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$  with  $\omega = v/R$ :

$$Mgh = \frac{1}{2}Mv^2(1 + c) \quad \Rightarrow \quad v = \sqrt{\frac{2gh}{1 + c}}.$$

**Execute** ( $h = 0.80$  m,  $2gh = 15.7$  m<sup>2</sup>/s<sup>2</sup>):

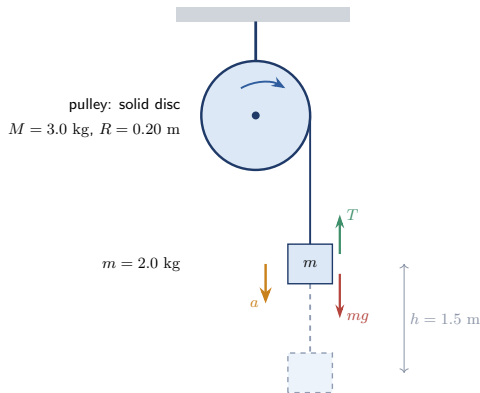
| Body                             | $c$  | $v$ at the bottom |
|----------------------------------|------|-------------------|
| frictionless block (no rotation) | 0    | 3.96 m/s          |
| solid sphere                     | 0.40 | 3.35 m/s          |
| solid disc                       | 0.50 | 3.23 m/s          |
| thin hoop                        | 1.00 | 2.80 m/s          |

**Evaluate:** the ranking is sphere, disc, hoop. The mass  $M$  and the radius  $R$  cancel, so only the shape ( $c$ ) matters. Every rolling body is slower than the sliding block because part of the energy goes into rotation. ✓

## Example 9: a mass on a cord turns a pulley

### Question

A 2.0 kg block hangs from a light cord wrapped round a solid disc pulley ( $M = 3.0$  kg,  $R = 0.20$  m) on a frictionless axle. The block is released from rest. Find (a) its acceleration and the tension, and (b) its speed after it has fallen 1.5 m, by energy.



### Predict first

Is  $a$  larger or smaller than  $g$ ? Is the tension greater than, equal to, or smaller than  $mg = 19.6$  N?

## The pulley: Newton's laws for the acceleration

**Set up:** downward positive for the block, the same sense of rotation for the pulley; the cord does not slip, so  $a = R\alpha$ .  $I = \frac{1}{2}MR^2$ .

## The pulley: Newton's laws for the acceleration

**Set up:** downward positive for the block, the same sense of rotation for the pulley; the cord does not slip, so  $a = R\alpha$ .  $I = \frac{1}{2}MR^2$ .

**Block:**  $mg - T = ma$ .     **Pulley:**  $TR = I\alpha = \frac{1}{2}MR^2\frac{a}{R} \Rightarrow T = \frac{1}{2}Ma$ .

## The pulley: Newton's laws for the acceleration

**Set up:** downward positive for the block, the same sense of rotation for the pulley; the cord does not slip, so  $a = R\alpha$ .  $I = \frac{1}{2}MR^2$ .

**Block:**  $mg - T = ma$ .     **Pulley:**  $TR = I\alpha = \frac{1}{2}MR^2 \frac{a}{R} \Rightarrow T = \frac{1}{2}Ma$ .

**Combine:**  $mg = (m + \frac{1}{2}M) a \Rightarrow a = \frac{mg}{m + M/2} = \frac{(2.0)(9.80)}{2.0 + 1.5} = 5.60 \text{ m/s}^2$ .

$T = \frac{1}{2}Ma = \frac{1}{2}(3.0)(5.60) = 8.40 \text{ N}$ .

## The pulley: Newton's laws for the acceleration

**Set up:** downward positive for the block, the same sense of rotation for the pulley; the cord does not slip, so  $a = R\alpha$ .  $I = \frac{1}{2}MR^2$ .

**Block:**  $mg - T = ma$ .     **Pulley:**  $TR = I\alpha = \frac{1}{2}MR^2 \frac{a}{R} \Rightarrow T = \frac{1}{2}Ma$ .

**Combine:**  $mg = (m + \frac{1}{2}M) a \Rightarrow a = \frac{mg}{m + M/2} = \frac{(2.0)(9.80)}{2.0 + 1.5} = 5.60 \text{ m/s}^2$ .

$T = \frac{1}{2}Ma = \frac{1}{2}(3.0)(5.60) = 8.40 \text{ N}$ .

**Evaluate:**  $a < g$  and  $T = 8.4 \text{ N} \ll mg = 19.6 \text{ N}$ : the pulley must be accelerated, so the tension is much less than the weight. (If  $T$  were  $mg$ , the block would not accelerate.) After falling 1.5 m:  $v = \sqrt{2ah} = \sqrt{2(5.60)(1.5)} = 4.10 \text{ m/s}$  and  $\omega = v/R = 20.5 \text{ rad/s}$ .

## The pulley: energy for the speed, and a check

**Set up:** system: block, pulley and Earth. The cord and the axle do no net work on the system, and  $\omega = v/R$ . State 1: at rest; state 2: after falling  $h = 1.5 \text{ m}$  ( $y = 0$  at state 2).

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**Execute:**  $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\frac{v^2}{R^2} = \frac{1}{2}\left(m + \frac{1}{2}M\right)v^2$ .

$$v = \sqrt{\frac{2mgh}{m + M/2}} = \sqrt{\frac{2(2.0)(9.80)(1.5)}{3.5}} = \sqrt{16.8} = 4.10 \text{ m/s}.$$

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**Evaluate:** the same 4.10 m/s as the force method, and smaller than the free-fall value  $\sqrt{2gh} = 5.42$  m/s because part of the energy goes into rotating the pulley:

$$\frac{1}{2}I\omega^2 = \frac{1}{2}(0.060)(20.5)^2 = 12.6 \text{ J of } 29.4 \text{ J. } \checkmark$$

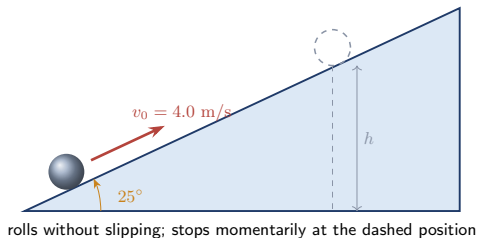
### Physical meaning

Energy gave  $v$  in one line;  $\sum \tau = I\alpha$  was needed for  $a$  and  $T$ . Choose by what is asked (Toolkit 6, step 5).

## Your turn: a sphere rolls up an incline

### Independent problem

A solid sphere rolls without slipping along a level floor at  $4.0 \text{ m/s}$  and then up a  $25^\circ$  incline. Find (a) the vertical height it reaches, and (b) the distance it travels along the incline before it stops. (c) How high would a frictionless block with the same initial speed go?



### Predict first

Does the sphere go higher or lower than the sliding block? Which has more kinetic energy at the start for the same speed?

**Set up:** state 1: on the floor at  $v = 4.0 \text{ m/s}$  and  $\omega = v/R$ ; state 2: at rest at height  $h$ .  
 $I = \frac{2}{5}MR^2$ . Static friction does no work.

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$$(a) \quad \frac{1}{2}Mv^2 + \frac{1}{2} \left( \frac{2}{5}MR^2 \right) \frac{v^2}{R^2} = \frac{7}{10}Mv^2 = Mgh \Rightarrow h = \frac{7v^2}{10g} = \frac{7(16.0)}{10(9.80)} = 1.14 \text{ m}.$$

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$$(b) \quad s = \frac{h}{\sin 25^\circ} = \frac{1.14}{0.4226} = 2.70 \text{ m}.$$

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$$(b) \quad s = \frac{h}{\sin 25^\circ} = \frac{1.14}{0.4226} = 2.70 \text{ m}.$$

$$(c) \quad \text{Sliding block: } h = \frac{v^2}{2g} = 0.816 \text{ m}.$$

**Evaluate:** the sphere goes higher (by the factor  $1.4 = 1 + c$ ) because its total kinetic energy at the same speed is 1.4 times larger.  $M$  and  $R$  cancel. ✓

## Momentum for rotation: angular momentum

For a particle with linear momentum  $\vec{p} = m\vec{v}$  (the product of mass and velocity) at position  $\vec{r}$  from a chosen point O:

### Rule: angular momentum

$$\vec{L} = \vec{r} \times \vec{p} \quad L = mvr \sin \varphi$$

For a rigid body turning about a fixed symmetry axis:  $L = I\omega$ ,  $[L] = \text{kg m}^2/\text{s}$ .

The direction of  $\vec{L}$  is given by the right-hand rule, along the axis for rotation about a symmetry axis.

### Rule: the rotational form of Newton's second law

$$\sum \vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}$$

### Physical meaning

The net external torque is the rate of change of angular momentum, as force is the rate of change of momentum. For fixed  $I$ :  $d(I\omega)/dt = I\alpha$ , the law of the previous contact.

## Concept check: angular momentum of a moving ball

### Question

A 0.50 kg ball moves in a straight line at 4.0 m/s, passing at a perpendicular distance of 1.2 m from a point O. (a) Find the magnitude of its angular momentum about O. (b) Does it change as the ball moves along its line? (c) What is  $L$  about a point on the ball's line?

### Predict first

Which quantity in  $mvr \sin \varphi$  changes as the ball moves along the line, and which stay fixed? Is there a torque about O?

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### Predict first

Which quantity in  $mvr \sin \varphi$  changes as the ball moves along the line, and which stay fixed? Is there a torque about O?

(a)  $L = mvr \sin \varphi = mvl = (0.50)(4.0)(1.2) = 2.4 \text{ kg m}^2/\text{s}$ , with  $l = r \sin \varphi$  the perpendicular distance. (b) No:  $l$  and  $v$  stay fixed (no force, so no torque about O). (c)  $L = 0$ , since  $l = 0$ .

**Angular momentum depends on the point chosen.**

# Conservation of angular momentum

## Rule: conservation of angular momentum

If the net external torque about an axis is zero, the angular momentum about that axis is constant:

$$\sum \tau_{\text{ext}} = 0 \Rightarrow L_1 = L_2 \quad \text{so} \quad I_1\omega_1 = I_2\omega_2$$

## Physical meaning

Internal forces cancel in pairs, so only external torques change  $L$ . The moment of inertia may change during the motion: if  $I$  falls,  $\omega$  must rise.

## Conditions

Gravity at the axis, a normal force through the axis and a hinge force at the axis give no torque about that axis. Conservation of  $L$  does *not* imply conservation of  $K$ .

## EQUATION – angular momentum

$$\vec{L} = \vec{r} \times \vec{p}, \quad L = I\omega, \quad \sum \vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}$$

**Meaning.** the quantity of rotational motion about a point; torque is its rate of change.

**Variables.**  $\vec{p} = m\vec{v}$ ;  $I$ ,  $\omega$  about the axis.

**Units.**  $\text{kg m}^2/\text{s}$ .

**Interpretation.** depends on the point chosen; a vector along the axis.

**Example.** the ball:  $L = 2.4 \text{ kg m}^2/\text{s}$ .

## EQUATION – conservation of $L$

$$\sum \tau_{\text{ext}} = 0 \Rightarrow I_1\omega_1 = I_2\omega_2$$

**Meaning.** without an external torque, the angular momentum does not change.

**Variables.**  $I_1$ ,  $\omega_1$  before;  $I_2$ ,  $\omega_2$  after.

**Units.**  $\text{kg m}^2/\text{s}$ .

**Interpretation.** if  $I$  falls,  $\omega$  rises;  $K$  is *not* conserved.

**Example.** the skater pulling in her arms.

### TOOLKIT 7 – Angular momentum and its conservation

- 1. SYSTEM AND AXIS:** choose the system and the axis, and the instants before (1) and after (2).
- 2. EXTERNAL TORQUES:** list the external forces; find their torques about the axis. If the net is zero (or the interaction is brief and the external torques are small),  $L$  is conserved.
- 3. EACH PART:**  $L = I\omega$  for a rotating body;  $L = mvl$  for a particle ( $l$  the perpendicular distance from the axis); give signs (anticlockwise positive).
- 4. WRITE**  $L_1 = L_2$  and solve.
- 5. ENERGY CHECK:** compare  $K_1$  and  $K_2$ : a sticking collision loses  $K$ ; a person pulling in the arms gains  $K$ .

**WATCH OUT:** using conservation of  $L$  when a net external torque acts; forgetting the signs of  $L$ ; assuming  $K$  is conserved when  $I$  changes.

**CHECK:** does  $\omega$  rise when  $I$  falls? Does the sign of  $K_2 - K_1$  agree with the physical process?

## Concept check: a skater pulls in her arms

### Answer, with a reason

A skater spinning with her arms out pulls them in. (a) What happens to her angular velocity? (b) To her angular momentum? (c) To her kinetic energy? (d) Where does any extra energy come from?

## Concept check: a skater pulls in her arms

### Answer, with a reason

A skater spinning with her arms out pulls them in. (a) What happens to her angular velocity? (b) To her angular momentum? (c) To her kinetic energy? (d) Where does any extra energy come from?

(a)  $\omega$  increases, because  $I$  decreases. (b)  $L$  is unchanged: the ice and gravity exert no net torque about her axis. (c)  $K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$  increases when  $I$  decreases. (d) From the chemical energy of her muscles: she does positive work pulling her arms inward.

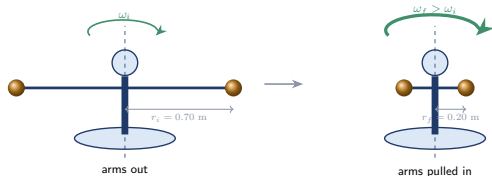
### Common error

“The skater turns faster, so energy is conserved.” Neither  $K$  nor the energy of the arms stays the same;  $L$  is the conserved quantity.

## Example 10: a person on a rotating stool

### Question

A person on a freely turning stool holds a  $3.0\text{ kg}$  mass in each hand and rotates at  $0.80\text{ rev/s}$  with the arms out ( $0.70\text{ m}$  from the axis). The person and stool alone have  $I = 2.5\text{ kg m}^2$ . The arms are pulled in to  $0.20\text{ m}$ . Find (a) the new rate of rotation, and (b) the kinetic energy before and after.



### Predict first

Will  $\omega$  at least double? By what factor does the kinetic energy change? Where does the extra energy come from?

## A person on a rotating stool: solution

**Set up:** no external torque about the vertical axis, so  $I_i\omega_i = I_f\omega_f$ . Treat the masses as point masses.

$$I_i = 2.5 + 2(3.0)(0.70)^2 = 5.44 \text{ kg m}^2; \quad I_f = 2.5 + 2(3.0)(0.20)^2 = 2.74 \text{ kg m}^2;$$

$$\omega_i = 0.80 \times 2\pi = 5.03 \text{ rad/s}.$$

## A person on a rotating stool: solution

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$$\omega_i = 0.80 \times 2\pi = 5.03 \text{ rad/s}.$$

$$\text{(a)} \quad \omega_f = \omega_i \frac{I_i}{I_f} = (5.03) \frac{5.44}{2.74} = 9.98 \text{ rad/s} = 1.59 \text{ rev/s}.$$

## A person on a rotating stool: solution

**Set up:** no external torque about the vertical axis, so  $I_i\omega_i = I_f\omega_f$ . Treat the masses as point masses.

$$I_i = 2.5 + 2(3.0)(0.70)^2 = 5.44 \text{ kg m}^2; \quad I_f = 2.5 + 2(3.0)(0.20)^2 = 2.74 \text{ kg m}^2;$$
$$\omega_i = 0.80 \times 2\pi = 5.03 \text{ rad/s}.$$

**(a)**  $\omega_f = \omega_i \frac{I_i}{I_f} = (5.03) \frac{5.44}{2.74} = 9.98 \text{ rad/s} = 1.59 \text{ rev/s}.$

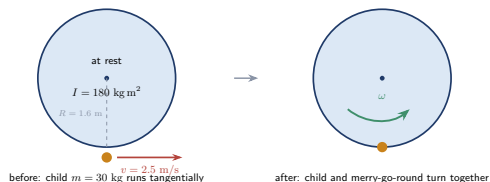
**(b)**  $K_i = \frac{1}{2}(5.44)(5.03)^2 = 68.7 \text{ J}; \quad K_f = \frac{1}{2}(2.74)(9.98)^2 = 136 \text{ J}.$

**Evaluate:**  $\omega$  rises by  $5.44/2.74 = 1.99$ , just short of doubling, and  $K$  rises by the same factor ( $K = \frac{1}{2}L\omega$  with  $L$  fixed). The extra 68 J is the work the person does pulling the masses inward; it is not conserved. ✓

## Your turn: a child jumps on a merry-go-round

### Independent problem

A 30 kg child runs at 2.5 m/s along a tangent to a stationary merry-go-round ( $I = 180 \text{ kg m}^2$ ,  $R = 1.6 \text{ m}$ ) and jumps on at the rim. Find (a) the angular momentum of the child about the axis before the jump, (b) the angular velocity afterwards, and (c) the fraction of the kinetic energy that is lost.



### Predict first

Is  $\omega$  larger or smaller than  $v/R = 1.56 \text{ rad/s}$ ? Is energy conserved in the jump?

**Set up:** the axle exerts no torque about the axis, and the jump is brief;  $L$  is conserved about the axis. The child is a particle with  $l = R$  (tangential motion).

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**(a)**  $L = mvR = (30)(2.5)(1.6) = 120 \text{ kg m}^2/\text{s}.$

**Set up:** the axle exerts no torque about the axis, and the jump is brief;  $L$  is conserved about the axis. The child is a particle with  $l = R$  (tangential motion).

(a)  $L = mvR = (30)(2.5)(1.6) = 120 \text{ kg m}^2/\text{s}.$

(b)  $I_{\text{tot}} = I + mR^2 = 180 + (30)(1.6)^2 = 257 \text{ kg m}^2$ , so  $\omega = \frac{L}{I_{\text{tot}}} = \frac{120}{257} = 0.467 \text{ rad/s}.$

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(b)  $I_{\text{tot}} = I + mR^2 = 180 + (30)(1.6)^2 = 257 \text{ kg m}^2$ , so  $\omega = \frac{L}{I_{\text{tot}}} = \frac{120}{257} = 0.467 \text{ rad/s}.$

(c)  $K_i = \frac{1}{2}(30)(2.5)^2 = 93.8 \text{ J}$ ;  $K_f = \frac{1}{2}(257)(0.467)^2 = 28.0 \text{ J}$ ; fraction lost  
 $= \frac{93.8 - 28.0}{93.8} = 0.70.$

**Evaluate:**  $\omega \ll v/R$ , as predicted, because the child shares  $L$  with a large  $I$ . 70% of the kinetic energy becomes thermal energy (the child's feet and the platform): a sticking collision does not conserve  $K$ . ✓

### Gravitation

Kepler's second law (equal areas in equal times) is conservation of angular momentum (supplementary notes).

### Lecture 11

Rotational kinetic energy  $\frac{1}{2}I\omega^2$  is the share of a diatomic molecule's energy that sets its heat capacity.

### Lecture 12

Engine power is torque times angular speed,  $P = \tau\omega$ ; efficiency follows from the second law.

### Physical meaning

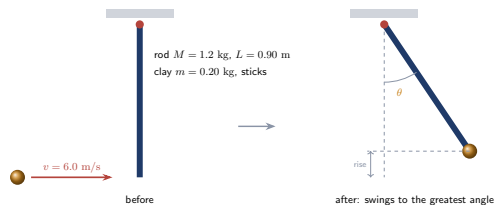
Rotation reuses every linear idea:  $F \rightarrow \tau$ ,  $m \rightarrow I$ ,  $p \rightarrow L$ ,  $W = Fx \rightarrow \tau\theta$ . The conservation laws of this course now come as a set.

## Integrated problem: clay strikes a hinged rod

**Decide what physics applies; no formula is given**

A 0.20 kg lump of clay moving horizontally at 6.0 m/s hits the lower end of a uniform rod ( $M = 1.2$  kg,  $L = 0.90$  m) hanging from a frictionless hinge at its top, and sticks.

- (a) Find the angular velocity just after the impact.
- (b) Find the greatest angle of swing.



**Predict first**

Which quantity is conserved in the impact, which in the swing? Is energy conserved in the impact?

**Identify:** two stages. *Impact:* short, with a large force from the hinge, but the hinge force has no torque about the hinge, and gravity has none either (it acts along the line through the hinge at that instant):  $L$  about the hinge is conserved. *Swing:* only gravity does work: energy is conserved.

**Identify:** two stages. *Impact:* short, with a large force from the hinge, but the hinge force has no torque about the hinge, and gravity has none either (it acts along the line through the hinge at that instant):  $L$  about the hinge is conserved. *Swing:* only gravity does work: energy is conserved.

**(a)**  $L_i = mvL = (0.20)(6.0)(0.90) = 1.08 \text{ kg m}^2/\text{s}.$

$$I = \frac{1}{3}ML^2 + mL^2 = 0.324 + 0.162 = 0.486 \text{ kg m}^2, \text{ so } \omega = \frac{1.08}{0.486} = 2.22 \text{ rad/s}.$$

**Identify:** two stages. *Impact:* short, with a large force from the hinge, but the hinge force has no torque about the hinge, and gravity has none either (it acts along the line through the hinge at that instant):  $L$  about the hinge is conserved. *Swing:* only gravity does work: energy is conserved.

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**(b)**  $K = \frac{1}{2}I\omega^2 = 1.20 \text{ J}.$  The centres of mass rise  $\frac{L}{2}(1 - \cos \theta)$  (rod) and  $L(1 - \cos \theta)$  (clay):  
 $1.20 = (Mg\frac{L}{2} + mgL)(1 - \cos \theta) = (5.29 + 1.76)(1 - \cos \theta) \Rightarrow \cos \theta = 0.830, \theta = 33.9^\circ.$

**Evaluate:** the clay had 3.6 J before and 1.2 J after: 2.4 J became thermal energy in the impact, so energy could not be used there; and  $L$  could not be used in the swing, because gravity exerts a net torque about the hinge. ✓

### Common mistakes and the correct habit

#### Mistake

#### Instead

Leaving the rotational term out of the energy equation.

$$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2.$$

Conserving  $L$  when an external torque acts, or assuming  $K$  is conserved.

List the external torques; compare  $K_1$  and  $K_2$ .

Forgetting that  $L$  depends on the point chosen.

Name the point or axis for every  $\vec{L}$  and  $\tau$ .

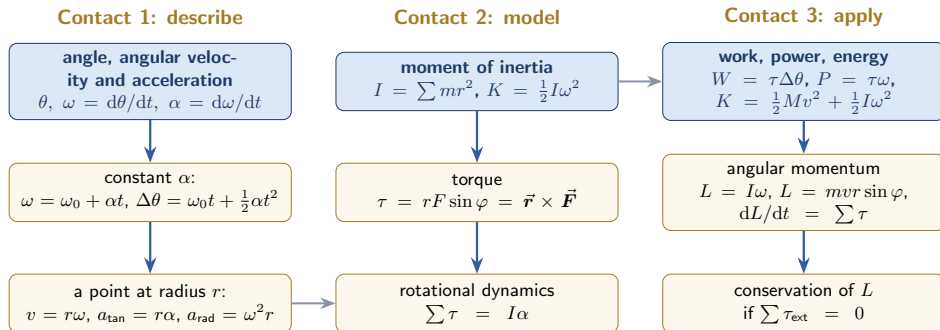
Saying static friction does work in rolling.

The contact point is momentarily at rest: no work.

Using  $v_{\text{cm}} = R\omega$  for a slipping wheel.

It holds only for rolling without slipping.

# Week 8 concept map



each idea has a linear partner:  $x, v, a \leftrightarrow \theta, \omega, \alpha$ ;  $m \leftrightarrow I$ ;  $F \leftrightarrow \tau$ ;  $p \leftrightarrow L$

## MASTER PROCEDURE – ROTATION

- 1. IDENTIFY** The body, the axis, and what is asked: an angle,  $\omega$ ,  $\alpha$ ,  $I$ , a torque, an energy, a power or an angular momentum.
  - 2. SKETCH** The axis, the positive sense, every force with its point of application, the states before and after.
  - 3. MODEL** Kinematics,  $\sum \tau = I\alpha$ , energy and work, or angular momentum; assumptions (rigid body, fixed axis, no slipping).
  - 4. PREDICT** Signs, direction, whether  $\omega$  rises or falls, whether  $K$  rises or falls, and the rough size.
  - 5. SELECT THE METHOD**  $\sum \tau = I\alpha$  for accelerations and forces; energy for speeds between two states;  $L$  when no external torque acts.
  - 6. EXECUTE** Radians throughout;  $\tau$  and  $I$  about the same axis (Toolkits 1 to 7).
  - 7. EVALUATE** Units, sign, magnitude, limiting cases, and a second method.
- SKETCH  $\rightarrow$  PREDICT  $\rightarrow$  CALCULATE  $\rightarrow$  CHECK.

## Inside the course framework

IDENTIFY  $\leftrightarrow$  step 1  
SET UP  $\leftrightarrow$  steps 2–5  
EXECUTE  $\leftrightarrow$  step 6  
EVALUATE  $\leftrightarrow$  step 7

## Physical meaning

Next: the Gravitation lecture. Kepler's second law (equal areas in equal times) is conservation of angular momentum for a planet.

## THE SEVEN TOOLKITS

- |   |                                                            |                                                                                                     |
|---|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1 | <b>Angular quantities and constant <math>\alpha</math></b> | radians; signs; $\omega = d\theta/dt$ ; missing variable                                            |
| 2 | <b>Linking linear and angular motion</b>                   | $v = r\omega$ , $a_{\text{tan}} = r\alpha$ , $a_{\text{rad}} = \omega^2 r$                          |
| 3 | <b>Moment of inertia</b>                                   | axis first; $\sum mr^2$ , table, parallel axis, composite                                           |
| 4 | <b>Torque</b>                                              | $rF \sin \varphi$ ; lever arm; sign; net torque                                                     |
| 5 | <b>Rotational dynamics</b>                                 | $\sum \tau = I\alpha$ ; free-body diagram of each body; $a = R\alpha$                               |
| 6 | <b>Work, power and energy in rotation</b>                  | $W = \tau \Delta\theta$ , $P = \tau\omega$ , $K = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$ ; rolling |
| 7 | <b>Angular momentum and its conservation</b>               | $L = I\omega$ , $L = mvl$ ; external torque; $K$ check                                              |

The Master Procedure sits above these. Week 1 to 5 toolkits (the cross product, components, free-body diagrams, constant acceleration, work and energy methods) are reused, not renamed.

### Key ideas

- Every point has the same  $\omega$  and  $\alpha$ ;  $v = r\omega$ .
- $I$  depends on the axis and on how the mass is spread.
- $\sum \tau = I\alpha$  is Newton's second law for rotation.
- Rolling bodies have translational and rotational  $K$ .
- With no external torque,  $L$  is conserved;  $K$  need not be.

### Where to look

**Definitions:** the vocabulary and equation-anatomy slides of each contact.

**Common mistakes:** the last slide of each contact.

**Connections:** the Connect boxes and slides.

## Summary: core equations

| Idea                                                   | Equation                                                                                     | Conditions and units               |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------|
| <b>Angle, <math>\omega</math>, <math>\alpha</math></b> | $\theta = s/r, \quad \omega = d\theta/dt, \quad \alpha = d\omega/dt$                         | rad, rad/s, rad/s <sup>2</sup>     |
| <b>Constant <math>\alpha</math></b>                    | $\omega = \omega_0 + \alpha t, \quad \theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2$ | only if $\alpha$ constant          |
| <b>Point on a body</b>                                 | $v = r\omega, \quad a_{\text{tan}} = r\alpha, \quad a_{\text{rad}} = \omega^2 r$             | radians                            |
| <b>Inertia</b>                                         | $I = \sum mr^2, \quad I = I_{\text{cm}} + Md^2$                                              | kg m <sup>2</sup>                  |
| <b>Torque</b>                                          | $\tau = rF \sin \varphi, \quad \sum \tau = I\alpha$                                          | same axis; N m                     |
| <b>Work, power</b>                                     | $W = \tau\Delta\theta, \quad P = \tau\omega$                                                 | J, W                               |
| <b>Energy</b>                                          | $K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2$                                     | rolling: $v_{\text{cm}} = R\omega$ |
| <b>Angular momentum</b>                                | $L = I\omega, \quad \sum \tau_{\text{ext}} = dL/dt$                                          | kg m <sup>2</sup> /s               |
| <b>Conservation</b>                                    | $I_1\omega_1 = I_2\omega_2$                                                                  | no external torque                 |

## Checklist: can I ...

- ☐ convert between degrees, radians, revolutions and rpm?
- ☐ find  $\omega$  and  $\alpha$  from  $\theta(t)$  and use the constant- $\alpha$  equations?
- ☐ relate  $v$ ,  $a_{\text{tan}}$  and  $a_{\text{rad}}$  to  $\omega$  and  $\alpha$ ?
- ☐ calculate  $I$  for particles, standard bodies and parallel axes?
- ☐ calculate a torque, with its sign, and a net torque?
- ☐ use  $\sum \tau = I\alpha$  with a cord, a pulley or a hinge?
- ☐ calculate the work done by a torque, the power and  $K = \frac{1}{2}I\omega^2$ ?
- ☐ use the energy equation with rotation, including rolling?
- ☐ state the condition for conservation of angular momentum?
- ☐ use  $L = I\omega$  and  $L = mvl$ , and explain what happens to  $K$ ?
- ☐ choose between  $\sum \tau = I\alpha$ , energy and  $L$  in an unfamiliar problem?

### After this week

Tutorial sheet: extra problems on every skill, with a challenge in which the method is not given. Supplementary notes: radians and units, the proofs of  $K = \frac{1}{2}I\omega^2$  and the parallel-axis theorem, the integrals for  $I$ , rolling dynamics, and Kepler's second law.